

A bank has two types of bank accounts: Current account and savings account.

a) Account information: account number and balance. For each account we can withdraw money from the account, as long as there was enough money, and deposit money.

b) Each account belongs to a client, and each client will be able to have up to ten different accounts. Client information: name, identity card and a vector of ten positions for saving her or his accounts.

c) The difference between the two accounts lies in the way of calculating interest:

► For *current account* the interest is worked out depending on annual interest rate.

► For *saving account*, if the money is lower than a minimum balance the interest is worked out using the half of annual interest rate, else the interest is worked out using the double of annual interest rate.

d) Draw the UML diagram. (optional)

e) Create the classes from the diagram and write the following methods:

- Constructors.
- getters and setters for each attribute.
- A method calculating the interest for each kind of account. Add the calculated interest to account balance.
- Methods for getting money out and in of the account.

f) Test the classes making a Java application. Create a vector for three clients with different accounts.

An example of toString() method in Client class:

```
Cliente: Jesús Manuel Rodríguez Segura-30117423B
Cuentas:
Numero de cuenta: 2222 Saldo: 0.0 Tipo: CuentaAhorro Saldo mínimo: 89.0
Numero de cuenta: 2221 Saldo: 3000.0 Tipo: CuentaCorriente
```